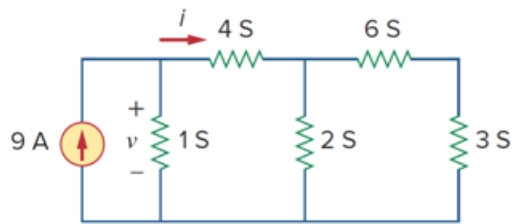


Problem

Obtain v and i in the circuit of Fig. 2.97.

Figure 2.97:



Step-by-step solution

Step 1 of 5

Refer to Figure 2.97 in the textbook.

From Figure 2.97 in the textbook, the values are given in conductance units.

The relation between resistance and conductance is,

$$\text{Resistance} \propto \frac{1}{\text{Conductance}}$$

Step 2 of 5

In Figure 2.97, the series combination of conductance 6 S and 3 S is in parallel with the conductance 2 S . Determine its equivalent conductance.

$$\begin{aligned} G_{\text{eq1}} &= 2 + \frac{(6)(3)}{6+3} \\ &= 2 + \frac{18}{9} \\ &= 2 + 2 \\ &= 4\text{ S} \end{aligned}$$

Now, draw the modified circuit diagram.

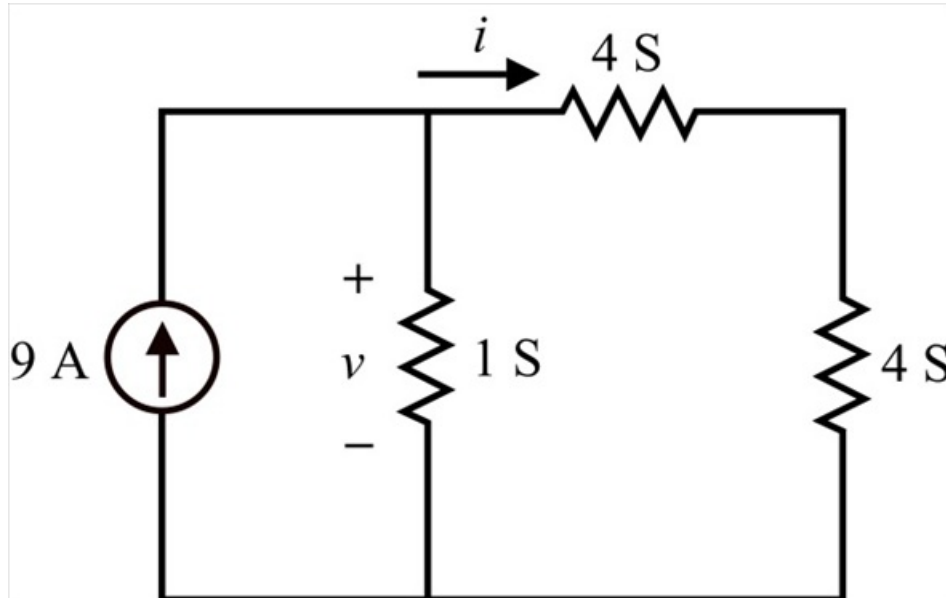


Figure 1

Step 3 of 5

In Figure 1, the two 4 S conductance's are in series, hence the same current, i will flow through both the conductance. Find its equivalent conductance and draw the modified diagram.

$$\begin{aligned} G_{\text{eq1}} &= \frac{(4)(4)}{4+4} \\ &= \frac{16}{8} \\ &= 2\text{ S} \end{aligned}$$

Next, draw the modified circuit diagram.

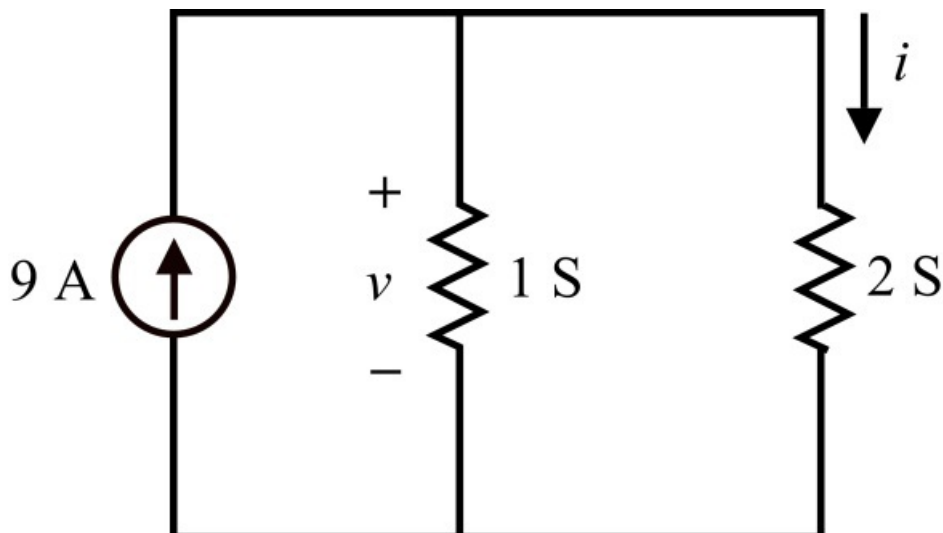


Figure 2

Step 4 of 5

Apply current division principle to obtain the current, i .

$$\begin{aligned} i &= \left(\frac{2}{1+2} \right) (9) \\ &= \left(\frac{2}{3} \right) (9) \\ &= 6 \text{ A} \end{aligned}$$

Therefore, the current, i is 6 A.

Step 5 of 5

Next, find the current flowing through the 1 S conductance. The current flowing through 1 S conductance is the difference between the total current and current flowing through 2 S conductance.

$$\begin{aligned} i_{1\text{S}} &= 9 - i \\ &= 9 - 6 \\ &= 3 \text{ A} \end{aligned}$$

Finally, obtain the voltage across the 1 S conductance.

$$\begin{aligned} v &= (3)(1) \\ &= 3 \text{ V} \end{aligned}$$

Therefore, the voltage, v is 3 V.